**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Defensive Design and Maintainability

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does authentication establish?

2. What is POP used for?

A) Sending new email
B) Transferring web pages
C) Looking up IP addresses
D) Downloading email

3. "Changing one protocol layer always requires every other layer to be redesigned." Correct this. Explain why.

4. A drive with private records is lost. "Encryption stops files being stolen or deleted." Correct this. What does it protect?

## Today's objectives

Explain defensive design and maintainable code. Apply this to another software scenario.

## Key vocabulary - Read, cover, recall

**Meaningful naming**

Descriptive identifiers such as totalCost make code easier to understand.

**Indentation**

Consistent indentation reveals which instructions belong in a block.

**Comments**

Explain important intent, rather than repeating obvious syntax line by line.

**Subprograms**

Splitting code into related functions or procedures helps reuse, testing and maintenance.

**Write one key term and its meaning in your own words:**

| Do Now | I Do | We Do | You Do | Review & Improve |
|---|---|---|---|---|
| **Recall previous learning** | Read new knowledge | Practise with guidance | Work independently | Check my understanding |

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Meaningful naming** | Descriptive identifiers such as totalCost make code easier to understand. Consider this in a library software. |
| **Indentation** | Consistent indentation reveals which instructions belong in a block. Consider this in a ticketing program. |
| **Comments** | Explain important intent, rather than repeating obvious syntax line by line. Consider this in a school marking tool. |
| **Subprograms** | Splitting code into related functions or procedures helps reuse, testing and maintenance. Consider this in a shop checkout. |

## I Do - One worked example

**A library software needs meaningful naming and indentation. Explain why. [4 marks]**

1. Identify the inputs, constraints and expected result.

2. Meaningful naming [1]; Descriptive identifiers such as totalCost make code easier to understand. [1].

3. Indentation [1]; Consistent indentation reveals which instructions belong in a block. [1].

4. Give a specific point and a linked reason for each feature.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a ticketing program, explain the purpose of meaningful naming. [2 marks]

2. A school marking tool needs indentation. Explain its role. [2 marks]

3. A shop checkout uses comments and subprograms. Explain both. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Add a comment explaining what the subtraction calculates.**

```
originalKB = 500
compressedKB = 300
// add your comment here
savedKB = originalKB - compressedKB
print(savedKB)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

## You Do - Independent exam questions

Q2. State TWO concepts relevant to this topic. [2 marks]

Q3. Explain how indentation and comments help in a travel app. [4 marks]

Q4. Explain the roles of meaningful naming, comments and subprograms in a club scoring. [6 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Descriptive identifiers such as totalCost make code easier to understand. ☐ T ☐ F
2. Descriptive identifiers such as totalCost make code easier to understand ☐ T ☐ F
3. Consistent indentation reveals which instructions belong in a block ☐ T ☐ F

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** Whether a user has supplied suitable evidence of the claimed identity. ☐

**Q2** D - Downloading email ☐

**Q3** Layers split communication into tasks. They provide services to each other and help compatibility, development and finding faults. ☐

**Q4** Without the key, an attacker cannot easily read ciphertext. Encryption does not stop theft or deletion. Use other controls for other threats. ☐

## Code First - Model

```
originalKB = 500
compressedKB = 300
// Work out how much storage compression saved.
savedKB = originalKB - compressedKB
print(savedKB)
```

**Output: 200**

## We Do - Check the model

1. Meaningful naming [1]; Descriptive identifiers such as totalCost make code easier to understand. [1]. ☐

2. Indentation [1]; Consistent indentation reveals which instructions belong in a block. [1]. ☐

3. Comments [1]; Explain important intent, rather than repeating obvious syntax line by line. [1]. Subprograms [1]; Splitting code into related functions or procedures helps reuse, testing and maintenance. [1]. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** Meaningful naming [1]; Comments [1]. ☐

**Q3** Indentation [1]; Consistent indentation reveals which instructions belong in a block. [1]. Comments [1]; Explain important intent, rather than repeating obvious syntax line by line. [1]. ☐

**Q4** Meaningful naming [1]; Descriptive identifiers such as totalCost make code easier to understand. [1]. Comments [1]; Explain important intent, rather than repeating obvious syntax line by line. [1]. Subprograms [1]; Splitting code into related functions or procedures helps reuse, testing and maintenance. [1]. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**